	D Y PATIL COLLEGE OF ENGINEERING AKURDI, PUNE APPROVED BY AICTE , RECOGNIZED BY DTE (GOVT.) AFFILIATED TO SAVITRIBAI PHULE PUNE UNIVERSITY Accredited by NAAC with 'A' Grade, NBA 6 UG Programs Accredited ISO 21001:2018 Certified Institute Autonomous Institute with effect from the Academic Year 2024-25 **Department of Computer Engineering**	

Document No: **Process No: DYP/Comp/**	**Unit Test 2 of BE** Subject Name**: Machine Learning(410242)**	**Academic Year: 2026-2027** **Sem I**
Rev: 00 Date:	Max. Marks**: 30** Duration**: 1 hour** Unit No.**: 3 and 4**	

SET 1

Instructions:

1. **Question 1 and 2 Compulsory.**
2. **Solve any three questions out of four (Question 1 and 2).**

Sr.no	Question Statement	CO	PO	PSO	BT level	marks
Q.1						
A.	Explain the concepts of Bias and Variance in machine learning. How do high bias and high variance affect model performance?	3	1, 2, 4	1	3	5
B.	Explain Linear Regression with its mathematical model. Describe how the best-fit regression line is obtained from training data.	3	1,3,4	1	2	5
C.	Differentiate between Ridge Regression and Lasso Regression based on regularization, penalty term, feature selection, and applications.	3	3,4,9	2	4	5
D.	Explain the following regression evaluation metrics with their formulas and significance: MAE, RMSE, and R^2 score	3	2,4,11	2	3	5
Q.2						
A.	Explain the K-Nearest Neighbour (KNN) classification algorithm. Describe the role of the value of K and explain how K affects classification performance.	4	1,2,3	1	3	5
B.	What is Ensemble Learning? Explain the basic idea of ensemble learning and differentiate between Bagging and Boosting.	4	3,4,5	1	4	5
C.	Explain Random Forest classification. Describe how multiple decision trees are constructed and how their predictions are combined.	4	3,4	2	2	5
D.	Explain the problems associated with balanced and imbalanced multiclass classification datasets. Why can accuracy be misleading for an imbalanced dataset?	4	4,5,11	2	4	5

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SET 2

Instructions:

1. **Question 1 and 2 Compulsory.**
2. **Solve any three questions out of four (Question 1 and 2).**

Sr. No	Question Statement	CO	PO	PSO	BT level	marks
	Q.1					
A.	What is Generalization in machine learning? Explain the relationship between generalization, underfitting, and overfitting with suitable examples.	3	1,2,3	1	3	5
B.	Differentiate between Underfitting and Overfitting. Explain at least three techniques used to overcome overfitting.	3	3,4	1	4	5
C.	Explain the Gradient Descent algorithm used in linear regression. Describe the role of learning rate and cost function.	3	1,2,3, 4	1	4	5
D.	Explain how L1 and L2 regularization help in controlling model complexity. Compare their effects on regression coefficients.	3	1,3,9, 10	2	4	5
	Q.2					
A.	Explain Support Vector Machine (SVM) with a suitable example. What are the concepts of hyperplane, support vectors, and margin?	4	1,3,4	1	3	5
B.	Differentiate between KNN and SVM with respect to working principle, training, computational complexity, sensitivity to features, and suitable applications.	4	4,5,1 1	1	4	5
C.	Explain AdaBoost algorithm with its working steps. How does AdaBoost give greater importance to incorrectly classified observations?	4	5,9	2	3	5
D.	Differentiate between Binary Classification and Multiclass Classification. Explain One-vs-All (OvA) and One-vs-One (OvO) approaches with suitable examples.	4	4,10, 11	2	5	5